



Tribhuvan University
Faculty of Humanities and Social Science
Semester: IV
Subject: Operating System
2021 Batch

Group B

Attempt any SIX questions

2. Define kernel. Explain different types of kernel in operating system.
3. What is PCB? What are different information that are included in a PCB?
4. How paging can be used for memory management? Also differentiate it with segmentation.
5. What are page faults and thrashing? Explain demand paging with protection bit.
6. How deadlock can be prevented? Explain in detail.
7. Explain different file access methods in detail.
8. What is distributed system? Explain the role of clock synchronization in distributed system.

Group C

Attempt any TWO questions.

9. What is critical section problem? How semaphores can be used to solve critical section problem Also explain producer consumer problem with its solution using Mutex.
10. What is belady's Anomaly in page replacement Algorithm? Explain with example. Consider the following page reference string:
7,2, 3, 1, 2, 5, 3, 4, 6, 7, 7, 1, 0, 5, 4, 6, 2, 3, 0, 1
How many page faults will occur if the program has four page frames available to it and uses the following page replacement algorithm?
 - a) FIFO replacement
 - b) LRU replacement
 - c) Optimal replacement

Don't Forget to Follow Uni Bytes

11. What is use of DMA controller? Consider the following process and answer the following questions.

Process	Allocation	Max	Available
	[A,B,C,D]	[A,B,C,D]	[A,B,C,D]
P0	0 0 1 1	0 0 1 2	1 5 2 0
P1	1 0 0 0	1 7 5 0	
P2	1 3 5 4	2 3 5 6	
P3	0 6 3 2	0 6 5 2	
P4	0 0 1 4	0 6 5 6	

- What is the content of need matrix?
- Is the system in safe state?
- If P2 request (3 4 0 2), can the request be granted Immediately.

Don't Forget to Follow Uni Bytes



Tribhuvan University
Faculty of Humanities and Social Science
Semester: IV
Subject: Operating System
2020 Batch

Group B

Attempt any SIX questions

2. Define Operating System? Explain objectives of Operating System.
3. How can we solve producer-consumer problem semaphore?
4. What do you mean by Preemptable and Non-Preemptable resources? Describe the necessary conditions for deadlock.
5. Given memory partitions of 200k, 600k, 300k, 400k and 700k (in order), how would each of the First-fit, Best-fit and Worst-fit algorithms place processes of 315k, 527k, 202k and 548k (in order)? Which algorithm makes the most efficient use of memory?
6. What is meant by system threat? Explain different types of program threat.
7. Explain methods of communication in a distributed system.
8. Write short notes on (Any Two):
 - a) Access Control Matrix
 - b) Process States
 - c) Kernel

Group C

Attempt any TWO questions.

9. The processes are assumed to have arrived in the order p1, p2, p3, p4, p5 all at time 0. Draw Gantt chart illustrating their execution and calculate average waiting time and turnaround time for the process given below using
 - a) First Come First Serve
 - b) Shortest Remaining Time Next
 - c) Priority
 - d) Round Robin (quantum = 2)

Don't Forget to Follow Uni Bytes

Process	Burst Time(sec)	Priority
P1	2	2
P2	1	1
P3	8	4
P4	4	2
P5	5	3

10. What is Disk Access Time and Disk Response Time? Suppose that a disk has 200 cylinders, numbered (0-199). The drive is currently serving a request at cylinder 43/and previous request was at cylinder 25. The queue of pending request, in FIFO order is: 55, 58, 39, 18, 90, 160, 150, 38, 184. Starting from the current head position, what is total distance (in cylinders) that the disk arm moves to satisfy all pending request for each of following disk scheduling algorithms?

a) FCFS b) SSTF c) C-SCAN d) LOOK

11. Differentiate between internal and external fragmentation? Explain briefly with examples and procedure to remove the types of fragmentation.

Don't Forget to Follow Uni Bytes



Tribhuvan University
Faculty of Humanities and Social Science
Semester: IV
Subject: Operating System
2019 Batch

Group B

Attempt any SIX questions

2. What is an operating system? Is "Multitasking" operating system better than "Batch" operating system? Justify your answer.
3. What is role of scheduler in operating system? Differentiate between short-term and long-term Scheduler.
4. What do you mean by Race condition? Consider two concurrently running process P1 with statement S1 and P2 with statement S2. Suppose we require that S2 be executed only after S1 has completed. For this synchronization need, write a semaphore solution.
5. What are CPU scheduling criteria's? Consider below table of four processes under multi-level queue scheduling. Queue number denotes queue of the process. Priority of queue 1 is greater than queue 2. Queue 1 use Round Robin with time quantum $t=2ms$ and queue 2 use FCFS. Draw Gantt chart to illustrate their execution and calculate average waiting and turnaround time also.
6. What is thrashing? Explain three memory management requirements?
7. Given five memory partition of 100KB, 500KB, 200KB, 300KB and 600KB (in order). How would the first fit, best fit and worst fit algorithm place processes of size 212KB, 417KB, 112KB, and 426KB (in order) in fixed-size partition? Which algorithm makes the most efficient use of memory?
8. When programmed I/O is suitable than other I/O handling techniques? Explain different type System Threat.

Don't Forget to Follow Uni Bytes

Group C

Attempt any TWO questions.

9. Consider the following System:

Process	Allocation	Max	Available
	[A,B,C,D]	[A,B,C,D]	[A,B,C,D]
P0	0 0 1 1	0 0 1 2	1 5 2 0
P1	1 0 0 0	1 7 5 0	
P2	1 3 5 4	2 3 5 6	
P3	0 6 3 2	0 6 5 2	
P4	0 0 1 4	0 6 5 6	

Answer the following questions using Banker's algorithm:

- What will be the contents of matrix need?
 - Draw the safe sequence?
 - If the request from P1 Arrives for (0, 4, 2, 0) can the request be granted immediately?
10. How second chance page replacement algorithm differ from FIFO page replacement policy?
Explain the concept of belay's anomaly with suitable example.
11. What is clock synchronization? Explain how physical clock is synchronized by Cristian algorithm and logical clock is synchronized by Lamport's algorithm with suitable example.

Don't Forget to Follow Uni Bytes.



Prepared by Uni Bytes / www.unibytes.xyz

Tribhuvan University
Faculty of Humanities and Social Science
Semester: IV
Subject: Operating System
2018 Batch

Group B

Attempt any SIX questions

2. What is an Operating System? Why is it known as the resource manager and the extended machine?
3. What is Kernel? What role does it play in an Operating System? Differentiate between monolithic and micro kernel.
4. Suppose that a disk has 200 cylinders (0-200). The head pointer is currently at 50 and previously at 35. The queue of pending request, in FIFO order is: 60, 190, 40, 125, 15, and 150. The time required to move per cylinder is 6 msec. starting from the current head position, what is the total seek time needed to satisfy all the pending request for each of the following disk scheduling algorithms?
a) FCFS b) SCAN
5. Define file and directory. Explain the concept of Access Control List (ACL) and Access Control Matrix (ACM).
6. What is biometric password in authentication? Explain the various system threats.
7. How distributed operating system is more applicable than centralized operating system? Explain the major goals of distributed operating system.
8. Write short notes on any TWO
a) Producer Consumer problem
b) Coalescing and Compaction
c) Ubuntu

Don't Forget to Follow Uni Bytes

Group C

Attempt any TWO questions.

9. What is CPU scheduling? Write down the criteria for CPU scheduling? Consider the following set of processes, with the length of the CPU burst given in milliseconds, draw Gantt chart illustrating their execution and calculate average waiting time and turnaround time using:

- a) First Come First Serve
- b) Shortest Job First
- c) Non-preemptive priority (smaller number implies higher priority)
- d) Round Robin (quantum=1)

Process	Burst Time(sec)	Priority
P0	10	3
P1	1	1
P2	2	5
P3	1	4
P4	5	2

The processes have arrived in the order P0, P1, P2, P3, P4 all at time 0.

10. Define the terms: page fault and thrashing. Consider the following page reference string: 1, 3, 5, 1, 7, 1, 5, 5, 1, 4, 3, 7, 6, 3, 4, 1. How many page faults would occur for each of the following page replacement algorithms assuming 3 page frames?

- a) FIFO page replacement
- b) LRU page replacement
- c) Optimal page replacement

11. Define deadlock. List out the conditions that can result in resource deadlock. Explain different deadlock handling methods in detail.

Don't Forget to Follow Uni Bytes



Tribhuvan University
Faculty of Humanities and Social Science
Semester: IV
Subject: Operating System
2017 Batch

Group B

Attempt any SIX questions

2. What is an operating system? Explain the functions of operating system.
3. Define the term semaphore. How does semaphore help in dining philosopher problem?
4. A system has two process and three resources. Each process needs a maximum of two resources. Is deadlock possible? Explain with answer.
5. Suppose a new process in a system arrives at an average of six processes per minute and each such process requires an average of 8 seconds of service time. Estimate the fraction of time the CPU is busy in a system with a single processor.
6. Given references to the following pages by a program,
0, 9, 0, 1, 8, 1, 8, 7, 8, 7, 1, 2, 8, 2, 7, 8, 2, 3, 8, 3
How many page faults will occur if the program has three page frames for each of the following algorithms?
 - a) FIFO
 - b) LRU
7. What is file? Explain how access control matrix provides resource protection that may access process.
8. What do you mean by one-time password in authentication? How worms are differing from virus.

Don't Forget to Follow Uni Bytes